a) $\nabla f \big|_{P_0} = 2xi + 2yj - 2zk \,(3,3,-4) = 6i + 10j + 8k$

Plano tang.

$6(x-3) + 10(y-5) + 8(z+4) = 0 \Rightarrow 6x - 18 + 10y - 50 + 8z + 32 = 0$

$\Rightarrow 6x + 10y + 8z = 36 \Rightarrow 3x + 5y + 4z = 18$

reta normal: $x = 3 + 6t$; $y = 5 + 10t$, $z = -4 + 8t$